SIAM Road Safety Hackathon

Aura-Node

Infrastructure-Independent V2V Safety Mesh & SOS Blackbox

Team: **Vision.exe**

Hack-4-Viksit Bharat 2026 Winner
India Innovates 2026 National 3rd rank

Priyanshu Chauhan (ECE-DU)
Shubham (ECE-DU)
Anshu Singh (ECE-DU)
Rank Aryan SIngh (ECE-DU)

The Problem

Visibility Collapse

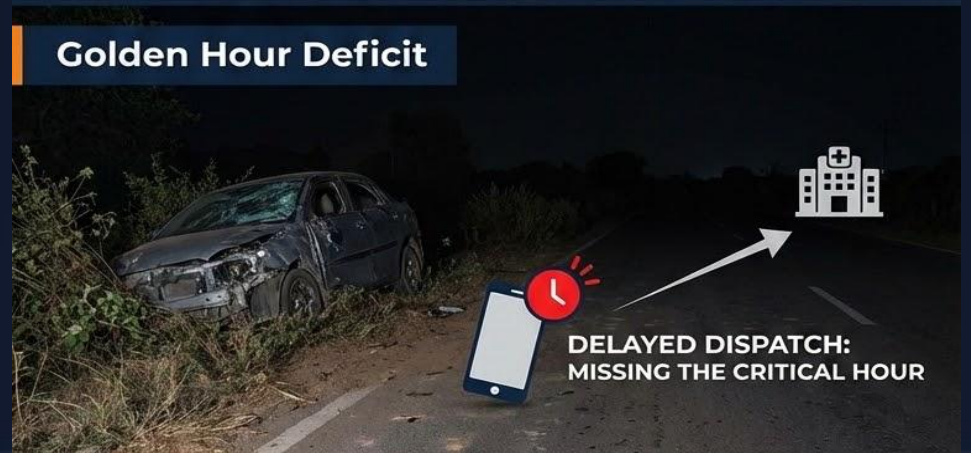
Optical sensors (Cameras/LiDAR) fail in Delhi's dense winter fog, rendering luxury ADAS useless and causing high-speed chain-reaction pileups.

Heterogeneous Reality

Over 70% of Indian traffic comprises two-wheelers. Standard car-based safety systems ignore the most vulnerable riders on the road.

Golden Hour Deficit

80% of severe fatalities occur when victims are hidden off-road. Reliance on bystanders delays dispatch, missing the critical Golden Hour.



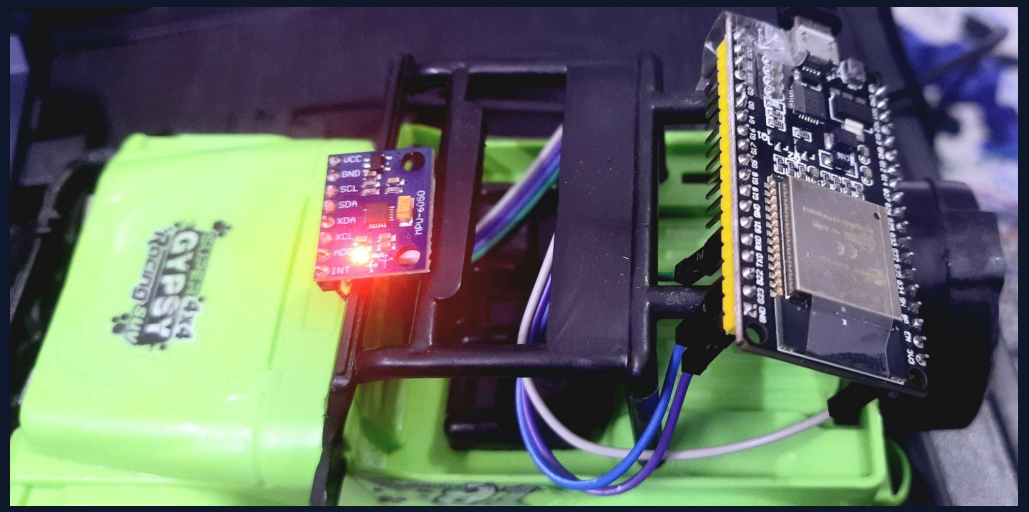
Our Solution: The Aura-Node

A retrofittable, clinical-grade safety module designed for autonomous collision avoidance and emergency intervention.

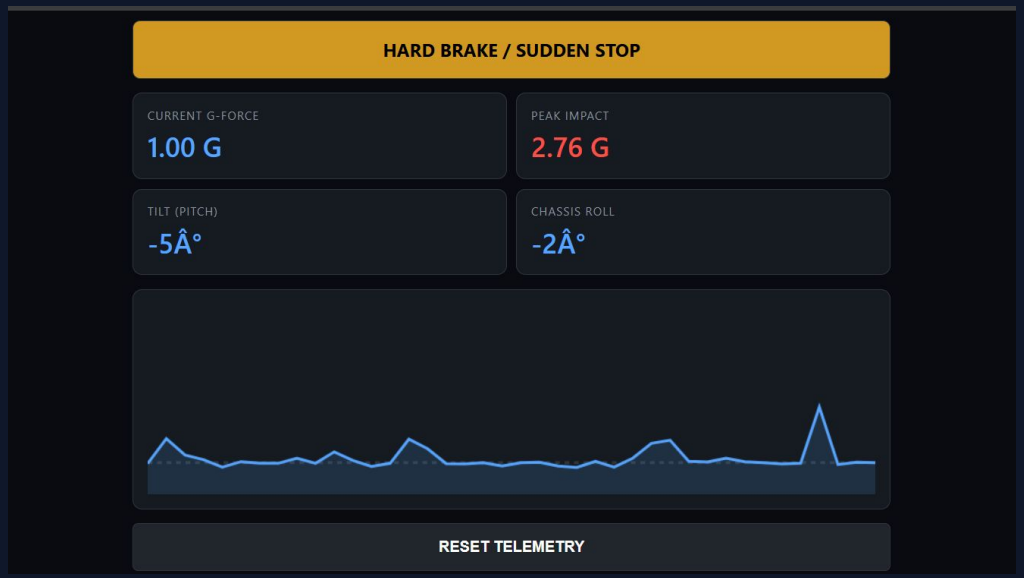
Preventative Fog Mesh
Utilizes Non-Line-of-Sight (NLOS) ESP-NOW protocol to broadcast vehicle telemetry to local cars with ultra-low 2-5ms latency.

Golden Hour SOS
An Edge-Kinematic engine continuously monitors 6-axis data to detect rollovers, instantly overriding the mesh to blast GPS coordinates.

Agnostic Retrofittability
Runs on a 3.7V LiPo or 12V Step-down. Protects a bicycle just as effectively as a commercial truck.



Prototype



Dashboard

Tech Stack & Architecture

Systems Design

- **Edge Hardware:** ESP32 Microcontroller, MPU6050 6-Axis IMU, NEO-6M GPS Module.
- **Comms Protocols:** ESP-NOW (Peer-to-peer MAC routing, zero latency), Store-and-Forward DTN.
- **Cloud Infrastructure:** Firebase NoSQL ledger & Twilio API for automated SMS/Voice escalation.

Algorithmic Anchor

To prevent false positives from severe Indian potholes, we calculate the continuous magnitude vector of acceleration:

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Paired with a software low-pass filter, this deterministic edge-computing algorithm triggers the Golden Hour SOS.

End-to-End System Pipeline

1. Edge Sensing

ESP32 + MPU6050 continuously sample 6-axis kinematics and highly precise GPS coordinates.

2. Edge Filtering

Math filter calculates absolute acceleration magnitude to isolate true crash events.

3. Mesh Ingestion

Decentralized ESP-NOW protocol routes hazard telemetry to nearby vehicles instantly (NLOS).

4. Cloud Ledger

Passing "Data Mules" reach cellular range and upload encrypted SOS payloads to GCP.

Feasibility, Challenges & Mitigation

Feasibility Factor	Key Challenges	Mitigation Strategy
Rapid Development: Utilizing the mature ESP32 C++ ecosystem ensures firmware prototyping within the hackathon window.	**Signal Noise (False Positives):** Severe Indian potholes can mimic the G-force of a crash, leading to SOS spam.	**Edge Math Filtering:** We process continuous acceleration magnitude with low-pass filters to ignore standard road bumps.
Low-Overhead Comms: ESP-NOW requires no external infrastructure, making immediate V2V deployment possible.	**Dead Zones:** Cellular towers do not cover 100% of national highways for cloud connectivity.	**Data Mules:** Utilizing passing vehicles as active nodes to securely carry the SOS packet out of deep dead zones.

National Impact & Economic Feasibility

Real-World Outcomes

Addressing the core metrics of the national road safety crisis through immediate, deployable intervention.

- **Zero-Latency Prevention:** Drastically reduces fatal multi-car pileups on winter highways by alerting drivers milliseconds before visual confirmation.
- **Golden Hour Rescue:** Guarantees automated emergency dispatch for hidden or off-road crashes, significantly increasing victim survival rates.
- **Inclusive Infrastructure:** Brings life-saving V2V communication to the 70% of Indian traffic totally ignored by standard luxury ADAS.

Feasibility & ROI

True infrastructure impact requires high node density. Aura-Node scales economically as a mass-adoption loss-leader.

Cost of a Pileup:

Asset Loss + Medical Liability = **₹50 Lakhs+**

Aura-Node BOM Cost:

ESP32 + IMU: ₹450

Power/PCB/Casing: ₹350

Total Node Cost: **~₹800**

Enterprise Integration & Future Scope

Commercial Overload

Using the gyro's pitch/roll data, edge firmware mathematically infers if a heavy truck is overloaded, warning the mesh of increased stopping distances.

Infrastructure Mapping

By aggregating Z-axis anomaly data, Aura-Nodes provide municipalities with real-time, zero-cost heat maps of highway potholes for predictive maintenance.

LoRaWAN Gateways

Upgrading toll-plazas and patrol vehicles with SX1278 LoRa modules to catch ESP-NOW SOS packets from data mules and blast them 10km to dispatch.

Zero Infrastructure.
Zero Latency.
Zero Casualties.

Aura-Node is ready to democratize road safety.

Thank You | Team Vision.exe